Lab 4

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns


sns.set(style="whitegrid")


pd.set_option("display.max_columns", None)

df = pd.read_csv("student_exam_performance.csv")
df.head(10)
```

```
   student_id  age  gender education_level school_type family_income  \
0  STU_000001   20  Female     High School      Public        Middle
1  STU_000002   17    Male     High School      Public        Middle
2  STU_000003   18   Other   Undergraduate      Public        Middle
3  STU_000004   20    Male     High School      Public        Middle
4  STU_000005   16  Female     High School     Private          High
5  STU_000006   18    Male     High School      Public          High
6  STU_000007   18    Male     High School     Charter        Middle
7  STU_000008   20  Female   Undergraduate     Private  Lower-Middle
8  STU_000009   15    Male   Undergraduate      Public  Lower-Middle
9  STU_000010   16  Female   Undergraduate      Public  Lower-Middle

  parent_education urban_rural  previous_exam_score  previous_gpa  \
0              NaN       Rural                78.36          2.95
1              NaN    Suburban                73.40          2.89
2           Master       Urban                82.76          3.35
3      High School    Suburban                60.61          2.57
4         Bachelor    Suburban                74.79          2.84
5      High School       Rural                67.68           NaN
6              NaN       Urban                78.84          3.00
7         Bachelor    Suburban                90.79          3.63
8        Associate    Suburban                61.28          2.43
9        Associate       Rural                61.33          2.72

   attendance_percentage  assignment_completion_rate class_participation  \
0                  81.49                       65.18              Medium
1                    NaN                       76.53                 Low
2                 100.00                       70.39              Medium
3                  84.91                       75.06                 Low
4                  88.21                       61.54              Medium
5                  97.11                       65.46              Medium
6                  85.38                       70.92              Medium
7                  80.61                       72.83                High
8                  86.47                       70.12                High
9                  89.61                       61.53                 Low

   study_hours_per_day  self_study_hours  private_tuition  \
0                 4.94              2.63                0
1                 3.42              2.64                0
2                 1.06              0.83                0
3                 7.89              5.51                0
4                 2.63              2.29                0
5                 3.18              1.67                1
6                 3.35              2.11                0
7                 5.51              4.33                0
8                 2.42              1.65                1
9                 1.26              0.78                0
```

	online_learning_hours	study_consistency	study_environment	study_method
0	1.75	Medium	Moderate	Flashcards
1	1.97	Medium	Moderate	Flashcards
2	3.40	Low	Moderate	Summarizing
3	1.94	Medium	Quiet	Summarizing
4	0.58	Medium	Noisy	Self-Reading
5	1.02	Medium	Quiet	Self-Reading
6	1.47	Medium	Quiet	Practice Tests
7	0.26	Low	Quiet	Self-Reading
8	6.23	High	Noisy	Practice Tests
9	2.86	Medium	Noisy	Group Study

	revision_frequency	practice_tests_completed	notes_quality	sleep_hours
0	Rarely	8	Average	6.31
1	Daily	10	Average	6.81
2	Weekly	5	Excellent	6.53
3	Weekly	8	Excellent	6.60
4	Rarely	1	Poor	7.83
5	Daily	6	Average	8.83
6	Weekly	6	Average	7.16
7	Weekly	9	Poor	8.22
8	Weekly	7	Excellent	6.55
9	Weekly	2	Poor	8.90

	sleep_quality	daily_screen_time	physical_activity_hours	break_frequency
0	Excellent	3.80	0.93	Frequently
1	Excellent	4.04	0.21	Frequently
2	Poor	3.19	2.56	Rarely
3	Fair	5.12	2.30	Rarely
4	Excellent	4.82	0.49	Frequently
5	Fair	8.37	1.64	Occasionally
6	Fair	2.59	0.71	Occasionally
7	Good	6.86	0.09	Frequently
8	Good	5.99	1.51	Occasionally
9	Good	1.21	1.59	Frequently

	stress_level	motivation_level	internet_access	device_availability
0	10	High	1	Shared
1	8	High	1	Shared
2	9	Medium	1	Dedicated
3	10	High	1	Dedicated
4	10	High	1	Dedicated
5	6	Medium	1	Shared
6	10	Medium	1	Dedicated
7	10	Medium	0	Shared
8	10	Medium	1	Dedicated
9	10	Medium	1	Dedicated

	educational_app_usage	online_course_hours	exam_difficulty
0	Low	2.15	Medium
1	Moderate	4.37	Easy
2	Low	1.43	Easy
3	Low	3.69	Medium
4	Moderate	1.97	Medium
5	Moderate	1.08	Hard
6	Low	5.62	Hard
7	Moderate	4.78	Easy
8	Low	2.01	Hard
9	Low	8.40	Hard

	exam_preparation_days	questions_attempted	questions_correct
0	27	95	86
1	4	100	86
2	28	96	74
3	16	97	77
4	28	98	52
5	27	96	65

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```
6                     12                  98                54
7                      7                  94                91
8                     14                  94                35
9                      4                  87                34

   time_management_score  exam_anxiety_level  exam_score performance_grade  \
0                    NaN                9.39       90.40                 A
1                  87.17                6.31       86.21                 A
2                  63.77                9.87       76.11                 B
3                  59.51                9.68       79.74                 B
4                  84.64               10.00       50.66                 D
5                  55.27                6.12       67.90                 C
6                  86.15                9.46       53.86                 D
7                  88.08               10.00       97.61                 A
8                  70.21                8.34       34.44                 F
9                  73.78                8.49       39.05                 F

  pass_status performance_level
0        Pass              High
1        Pass              High
2        Pass            Medium
3        Pass            Medium
4        Pass               Low
5        Pass            Medium
6        Pass               Low
7        Pass              High
8        Fail               Low
9        Fail               Low
```

```
df.dtypes
```

```
student_id                    object
age                            int64
gender                        object
education_level               object
school_type                   object
family_income                 object
parent_education              object
urban_rural                   object
previous_exam_score          float64
previous_gpa                 float64
attendance_percentage        float64
assignment_completion_rate   float64
class_participation           object
study_hours_per_day          float64
self_study_hours             float64
private_tuition                int64
online_learning_hours        float64
study_consistency             object
study_environment             object
study_method                  object
revision_frequency            object
practice_tests_completed       int64
notes_quality                 object
sleep_hours                  float64
sleep_quality                 object
daily_screen_time            float64
physical_activity_hours      float64
break_frequency               object
stress_level                   int64
motivation_level              object
internet_access                int64
device_availability           object
educational_app_usage         object
online_course_hours          float64
exam_difficulty               object
```

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```
exam_preparation_days           int64
questions_attempted             int64
questions_correct               int64
time_management_score         float64
exam_anxiety_level            float64
exam_score                    float64
performance_grade              object
pass_status                    object
performance_level              object
dtype: object
```

```
df.isnull().sum()
```

```
student_id                       0
age                              0
gender                           0
education_level                  0
school_type                      0
family_income                    0
parent_education              6526
urban_rural                      0
previous_exam_score              0
previous_gpa                  7793
attendance_percentage         9863
assignment_completion_rate       0
class_participation              0
study_hours_per_day              0
self_study_hours                 0
private_tuition                  0
online_learning_hours            0
study_consistency                0
study_environment                0
study_method                     0
revision_frequency               0
practice_tests_completed         0
notes_quality                 8407
sleep_hours                      0
sleep_quality                 6994
daily_screen_time                0
physical_activity_hours          0
break_frequency                  0
stress_level                     0
motivation_level                 0
internet_access                  0
device_availability           2965
educational_app_usage            0
online_course_hours              0
exam_difficulty                  0
exam_preparation_days            0
questions_attempted              0
questions_correct                0
time_management_score         9613
exam_anxiety_level               0
exam_score                       0
performance_grade                0
pass_status                      0
performance_level                0
dtype: int64
```

```
df_removed = df_missing.dropna()
df_removed.shape
```

```
(58060, 44)
```

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```
df_missing.head(10)
```

	student_id	age	gender	education_level	school_type	family_income	parent_education	urban_rural	previous_exam_score	previous_gpa	attendance_percentage	assignment_completion_rate	class_participation	study_hours_per_day	self_study_hours	private_tuition	online_learning_hours	study_consistency	study_environment	study_method	revision_frequency	practice_tests_completed	notes_quality	sleep_hours
0	STU_000001	20	Female	High School	Public	Middle	NaN	Rural	78.36	2.95	81.49	65.18	Medium	4.94	2.63	0	1.75	Medium	Moderate	Flashcards	Rarely	8	Average	6.31
1	STU_000002	17	Male	High School	Public	Middle	NaN	Suburban	73.40	2.89	NaN	76.53	Low	3.42	2.64	0	1.97	Medium	Moderate	Flashcards	Daily	10	Average	6.81
2	STU_000003	18	Other	Undergraduate	Public	Middle	Master	Urban	82.76	3.35	100.00	70.39	Medium	1.06	0.83	0	3.40	Low	Moderate	Summarizing	Weekly	5	Excellent	6.53
3	STU_000004	20	Male	High School	Public	Middle	High School	Suburban	60.61	2.57	84.91	75.06	Low	7.89	5.51	0	1.94	Medium	Quiet	Summarizing	Weekly	8	Excellent	6.60
4	STU_000005	16	Female	High School	Private	High	Bachelor	Suburban	74.79	2.84	88.21	61.54	Medium	2.63	2.29	0	0.58	Medium	Noisy	Self-Reading	Rarely	8	Average	6.31
5	STU_000006	18	Male	High School	Public	High	High School	Rural	67.68	NaN	97.11	65.46	Medium	3.18	1.67	1	1.02	Medium	Quiet	Self-Reading	Daily	10	Average	6.81
6	STU_000007	18	Male	High School	Charter	Middle	NaN	Urban	78.84	3.00	85.38	70.92	Medium	3.35	2.11	0	1.47	Medium	Quiet	Practice Tests	Weekly	5	Excellent	6.53
7	STU_000008	20	Female	Undergraduate	Private	Lower-Middle	Bachelor	Suburban	90.79	3.63	80.61	72.83	High	5.51	4.33	0	0.26	Low	Quiet	Self-Reading	Weekly	8	Excellent	6.60
8	STU_000009	15	Male	Undergraduate	Public	Lower-Middle	Associate	Suburban	61.28	2.43	86.47	70.12	High	2.42	1.65	1	6.23	High	Noisy	Practice Tests	Rarely	8	Average	6.31
9	STU_000010	16	Female	Undergraduate	Public	Lower-Middle	Associate	Rural	61.33	2.72	89.61	61.53	Low	1.26	0.78	0	2.86	Medium	Noisy	Group Study	Daily	10	Average	6.81

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```
4             Rarely                         1          Poor         7.83
5              Daily                         6       Average         8.83
6             Weekly                         6       Average         7.16
7             Weekly                         9          Poor         8.22
8             Weekly                         7     Excellent         6.55
9             Weekly                         2          Poor         8.90

  sleep_quality  daily_screen_time  physical_activity_hours break_frequency  \
0     Excellent               3.80                     0.93      Frequently
1     Excellent               4.04                     0.21      Frequently
2          Poor               3.19                     2.56          Rarely
3          Fair               5.12                     2.30          Rarely
4     Excellent               4.82                     0.49      Frequently
5          Fair               8.37                     1.64    Occasionally
6          Fair               2.59                     0.71    Occasionally
7          Good               6.86                     0.09      Frequently
8          Good               5.99                     1.51    Occasionally
9          Good               1.21                     1.59      Frequently

   stress_level motivation_level  internet_access device_availability  \
0            10             High                1              Shared
1             8             High                1              Shared
2             9           Medium                1           Dedicated
3            10             High                1           Dedicated
4            10             High                1           Dedicated
5             6           Medium                1              Shared
6            10           Medium                1           Dedicated
7            10           Medium                0              Shared
8            10           Medium                1           Dedicated
9            10           Medium                1           Dedicated

  educational_app_usage  online_course_hours exam_difficulty  \
0                   Low                 2.15           Medium
1              Moderate                 4.37             Easy
2                   Low                 1.43             Easy
3                   Low                 3.69           Medium
4              Moderate                 1.97           Medium
5              Moderate                 1.08             Hard
6                   Low                 5.62             Hard
7              Moderate                 4.78             Easy
8                   Low                 2.01             Hard
9                   Low                 8.40             Hard

   exam_preparation_days  questions_attempted  questions_correct  \
0                     27                   95                 86
1                      4                  100                 86
2                     28                   96                 74
3                     16                   97                 77
4                     28                   98                 52
5                     27                   96                 65
6                     12                   98                 54
7                      7                   94                 91
8                     14                   94                 35
9                      4                   87                 34

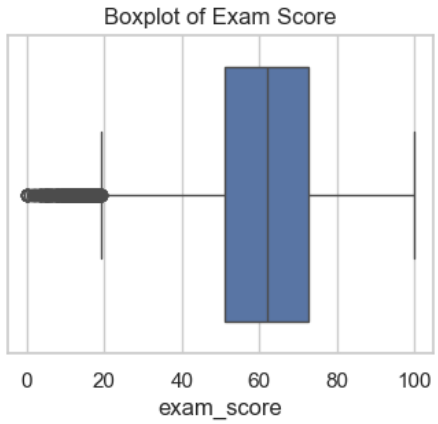
   time_management_score  exam_anxiety_level  exam_score performance_grade  \
0                    NaN                9.39       90.40                 A
1                  87.17                6.31       86.21                 A
2                  63.77                9.87       76.11                 B
3                  59.51                9.68       79.74                 B
4                  84.64               10.00       50.66                 D
5                  55.27                6.12       67.90                 C
6                  86.15                9.46       53.86                 D
7                  88.08               10.00       97.61                 A
8                  70.21                8.34       34.44                 F
9                  73.78                8.49       39.05                 F

  pass_status performance_level
```

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```
0        Pass              High
1        Pass              High
2        Pass            Medium
3        Pass            Medium
4        Pass               Low
5        Pass            Medium
6        Pass               Low
7        Pass              High
8        Fail               Low
9        Fail               Low
```

```
plt.figure(figsize=(4, 3))
sns.boxplot(x=df['exam_score'])
plt.title("Boxplot of Exam Score")
plt.show()
```




```
Q1 = df['exam_score'].quantile(0.25)
Q3 = df['exam_score'].quantile(0.75)
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[(df['exam_score'] < lower) | (df['exam_score'] > upper)]
outliers.head(15)
```

```
      student_id  age  gender education_level school_type family_income  \
78    STU_000079   17    Male   Undergraduate      Public        Middle
184   STU_000185   20    Male     High School      Public        Middle
194   STU_000195   16    Male   Undergraduate     Private  Lower-Middle
308   STU_000309   14    Male     High School     Charter  Lower-Middle
500   STU_000501   20    Male     High School     Private  Lower-Middle
880   STU_000881   19    Male     High School      Public        Middle
923   STU_000924   18  Female     High School     Private  Upper-Middle
1319  STU_001320   17  Female   Undergraduate     Private           Low
1720  STU_001721   16    Male     High School     Private  Lower-Middle
1781  STU_001782   15    Male     High School     Private        Middle
1804  STU_001805   20  Female   Undergraduate     Private           Low
2071  STU_002072   15  Female   Undergraduate     Private           Low
2199  STU_002200   19  Female     High School     Private           Low
3104  STU_003105   14  Female     High School      Public           Low
3278  STU_003279   19    Male   Undergraduate      Public  Lower-Middle

     parent_education urban_rural  previous_exam_score  previous_gpa  \
78           Associate       Urban                56.96          1.93
184       High School    Suburban                67.69          2.98
194       High School    Suburban                46.21          1.78
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308	High School	Suburban	66.21	2.49
500	High School	Urban	62.67	2.27
880	Bachelor	Suburban	63.82	2.53
923	Master	Suburban	48.41	1.94
1319	Bachelor	Rural	63.30	2.48
1720	Bachelor	Rural	56.75	2.21
1781	Doctorate	Urban	51.52	1.99
1804	Associate	Suburban	53.28	1.64
2071	Associate	Suburban	66.82	2.71
2199	Master	Urban	53.17	2.04
3104	High School	Urban	61.57	2.45
3278	NaN	Suburban	56.52	2.28

	attendance_percentage	assignment_completion_rate	class_participation
78	71.44	48.62	Medium
184	87.06	66.98	Medium
194	81.87	81.63	High
308	74.52	56.96	High
500	86.46	58.81	Medium
880	97.43	85.80	Medium
923	86.52	65.30	High
1319	80.42	84.97	High
1720	NaN	62.92	Medium
1781	81.30	79.58	Medium
1804	74.94	69.18	Medium
2071	78.83	38.37	High
2199	79.24	61.79	High
3104	96.82	61.92	Medium
3278	75.53	52.24	Medium

	study_hours_per_day	self_study_hours	private_tuition
78	0.96	0.60	0
184	1.52	1.29	0
194	3.42	2.61	1
308	1.62	1.31	1
500	0.50	0.39	0
880	1.65	0.92	0
923	1.48	0.86	0
1319	2.45	1.96	1
1720	2.94	2.60	0
1781	0.59	0.52	1
1804	1.18	0.78	0
2071	0.60	0.33	0
2199	2.10	1.22	1
3104	0.90	0.78	0
3278	0.95	0.51	0

	online_learning_hours	study_consistency	study_environment
78	5.79	Medium	Noisy
184	2.50	Low	Noisy
194	1.35	Medium	Moderate
308	0.84	High	Moderate
500	3.81	High	Moderate
880	8.83	Medium	Quiet
923	1.60	High	Moderate
1319	3.15	Medium	Moderate
1720	3.90	High	Quiet
1781	0.37	Low	Quiet
1804	0.66	Medium	Moderate
2071	0.99	Medium	Moderate
2199	1.12	High	Moderate
3104	0.13	Medium	Moderate
3278	0.83	Medium	Noisy

	study_method	revision_frequency	practice_tests_completed
78	Group Study	Weekly	3
184	Group Study	Weekly	1
194	Self-Reading	Weekly	4

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308	Flashcards	Weekly	3
500	Group Study	Weekly	4
880	Practice Tests	Rarely	0
923	Flashcards	Rarely	1
1319	Flashcards	Weekly	4
1720	Practice Tests	Weekly	3
1781	Summarizing	Rarely	2
1804	Flashcards	Weekly	2
2071	Practice Tests	Rarely	4
2199	Practice Tests	Weekly	5
3104	Self-Reading	Weekly	5
3278	Self-Reading	Daily	4

	notes_quality	sleep_hours	sleep_quality	daily_screen_time
78	Poor	6.42	Good	4.43
184	Poor	6.17	Excellent	5.45
194	Average	5.53	Excellent	4.81
308	NaN	4.30	NaN	6.72
500	Poor	6.54	Good	3.42
880	Average	6.19	Good	5.90
923	Average	7.93	Good	4.91
1319	Poor	6.95	Fair	6.55
1720	Excellent	7.65	Good	3.78
1781	Average	6.79	Good	4.56
1804	Excellent	7.39	Fair	3.83
2071	Excellent	5.23	Excellent	5.47
2199	Average	6.44	Fair	6.11
3104	Excellent	6.48	Fair	2.91
3278	Average	6.57	Fair	5.44

	physical_activity_hours	break_frequency	stress_level	motivation_level
78	0.75	Rarely	10	Medium
184	5.71	Occasionally	10	High
194	1.66	Rarely	10	Low
308	2.60	Occasionally	9	Medium
500	1.93	Rarely	10	Medium
880	0.60	Rarely	10	Medium
923	0.88	Occasionally	10	Medium
1319	0.79	Occasionally	10	Medium
1720	2.99	Frequently	10	High
1781	6.33	Frequently	10	Medium
1804	0.61	Frequently	10	Medium
2071	0.48	Rarely	10	Medium
2199	0.81	Occasionally	10	Medium
3104	0.13	Rarely	7	High
3278	6.42	Rarely	9	High

	internet_access	device_availability	educational_app_usage
78	1	Dedicated	Low
184	1	Dedicated	Low
194	1	Shared	Low
308	0	Shared	Low
500	1	Dedicated	Low
880	1	Dedicated	Low
923	1	Dedicated	High
1319	1	Dedicated	Low
1720	1	Shared	Low
1781	1	Dedicated	High
1804	1	Dedicated	High
2071	1	Dedicated	High
2199	0	Dedicated	Moderate
3104	1	Dedicated	Moderate
3278	0	Dedicated	Low

	online_course_hours	exam_difficulty	exam_preparation_days
78	6.17	Hard	2
184	9.88	Hard	1
194	1.77	Hard	2

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```
308                  0.32           Hard                     1
500                  2.67           Hard                     1
880                  0.91           Hard                     1
923                  1.64         Medium                     2
1319                 1.38           Hard                     1
1720                 4.83           Hard                     2
1781                 5.48           Easy                     1
1804                 0.25           Easy                     2
2071                 6.94           Easy                     1
2199                 1.81           Hard                    29
3104                 0.19           Hard                     1
3278                 0.71           Hard                     2

      questions_attempted  questions_correct  time_management_score  \
78                     89                  9                  75.76
184                    93                 13                  69.67
194                    93                 10                  59.57
308                    96                 16                  70.33
500                    85                  2                    NaN
880                    98                 14                  75.50
923                    88                 11                  96.61
1319                   95                 19                  72.89
1720                   94                 14                  81.25
1781                   89                  8                  80.48
1804                   89                 13                  73.18
2071                   89                 14                  56.05
2199                   94                  7                  96.78
3104                   88                 11                  74.60
3278                  100                 17                  61.55

      exam_anxiety_level  exam_score performance_grade pass_status  \
78                  8.89        9.60                 F        Fail
184                 8.96       12.56                 F        Fail
194                10.00        9.60                 F        Fail
308                 8.95       12.74                 F        Fail
500                 9.73        1.90                 F        Fail
880                 9.71       13.65                 F        Fail
923                 9.73        9.43                 F        Fail
1319                9.57       17.60                 F        Fail
1720                8.58       14.24                 F        Fail
1781                9.09        9.19                 F        Fail
1804               10.00       16.75                 F        Fail
2071                9.71       16.93                 F        Fail
2199               10.00        9.73                 F        Fail
3104                7.42       13.37                 F        Fail
3278               10.00       17.95                 F        Fail

     performance_level
78                 Low
184                Low
194                Low
308                Low
500                Low
880                Low
923                Low
1319               Low
1720               Low
1781               Low
1804               Low
2071               Low
2199               Low
3104               Low
3278               Low


df_no_outliers = df[(df['exam_score'] >= lower) & (df['exam_score'] <= upper)]

print("Original shape: ", df.shape)
```

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```
print("After removing outliers: ", df_no_outliers.shape)
```

Original shape: (100000, 44)
After removing outliers: (99627, 44)

```
df[['exam_score', 'performance_grade']].head()
```

```
   exam_score performance_grade
0       90.40                 A
1       86.21                 A
2       76.11                 B
3       79.74                 B
4       50.66                 D
```

```
import numpy as np

cols_to_scale = ['previous_exam_score', 'previous_gpa', 'attendance_percentage',
                 'study_hours_per_day', 'exam_anxiety_level', 'exam_score']

df_standardized = df[cols_to_scale].copy()
df_standardized = (df_standardized - df_standardized.mean()) / df_standardized.std(ddof=0)

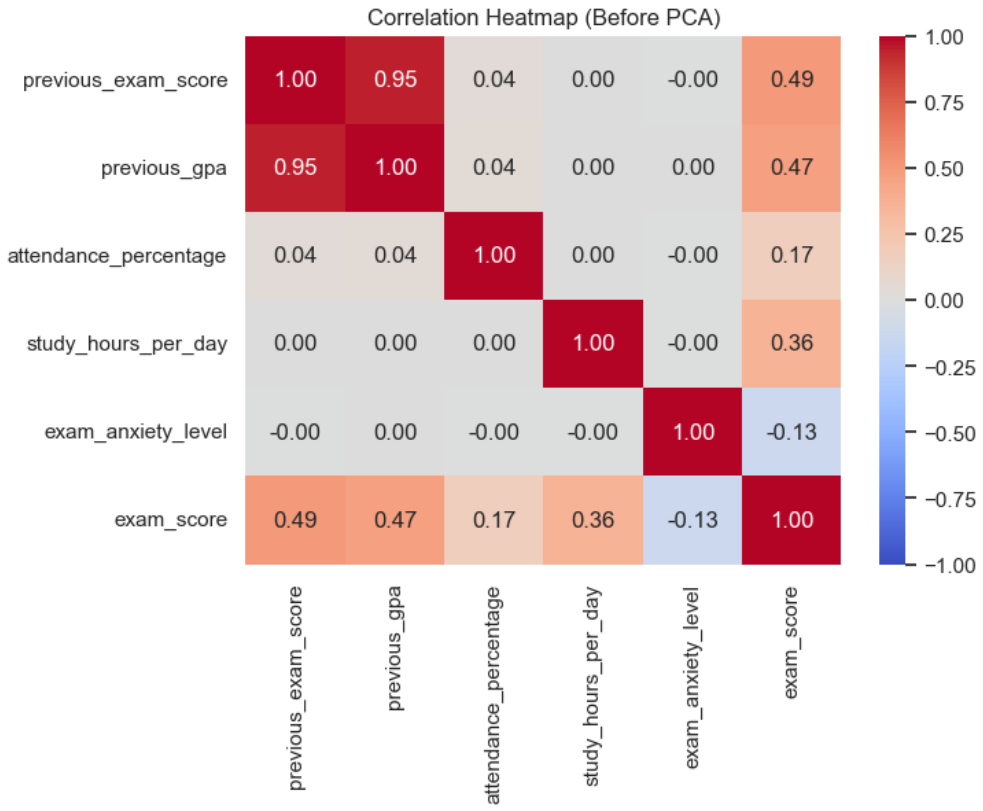
df_standardized.head()
```

```
   previous_exam_score  previous_gpa  attendance_percentage  \
0             0.775897      0.353623              -0.394687
1             0.340008      0.228157                    NaN
2             1.162572      1.190067               1.941714
3            -0.783987     -0.440998               0.036998
4             0.462162      0.123601               0.453537

   study_hours_per_day  exam_anxiety_level  exam_score
0             1.130750            0.341456    1.793413
1             0.246750           -1.931829    1.529286
2            -1.125777            0.695734    0.892607
3             2.846407            0.555499    1.121433
4            -0.212698            0.791684   -0.711698
```

```
plt.figure(figsize=(7, 5))
sns.heatmap(df_standardized[cols_to_scale].corr(),
            annot=True, cmap="coolwarm", fmt=".2f", vmin=-1, vmax=1)
plt.title("Correlation Heatmap (Before PCA)")
plt.show()
```

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